

FRANK PORTER
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M1 MACRO

Group A

Session 17

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

Model

Firms

$$y_t = z_t h_t$$

↑
output

↙

labor (Hours worked)

↑
Exogenous Stochastic Productivity

Every period:

(Competition)

$$\left[\begin{array}{l} \text{Max}_{(y_t, h_t)} \Pi_t = P_t y_t - w_t h_t \\ \text{s.t.} \quad y_t = z_t h_t \end{array} \right.$$

$$\text{Net } \pi_t = p_t z_t h_t - w_t h_t$$

$$[\neq y_t = z_t p_t^\alpha]$$

$$0 < \alpha < 1$$

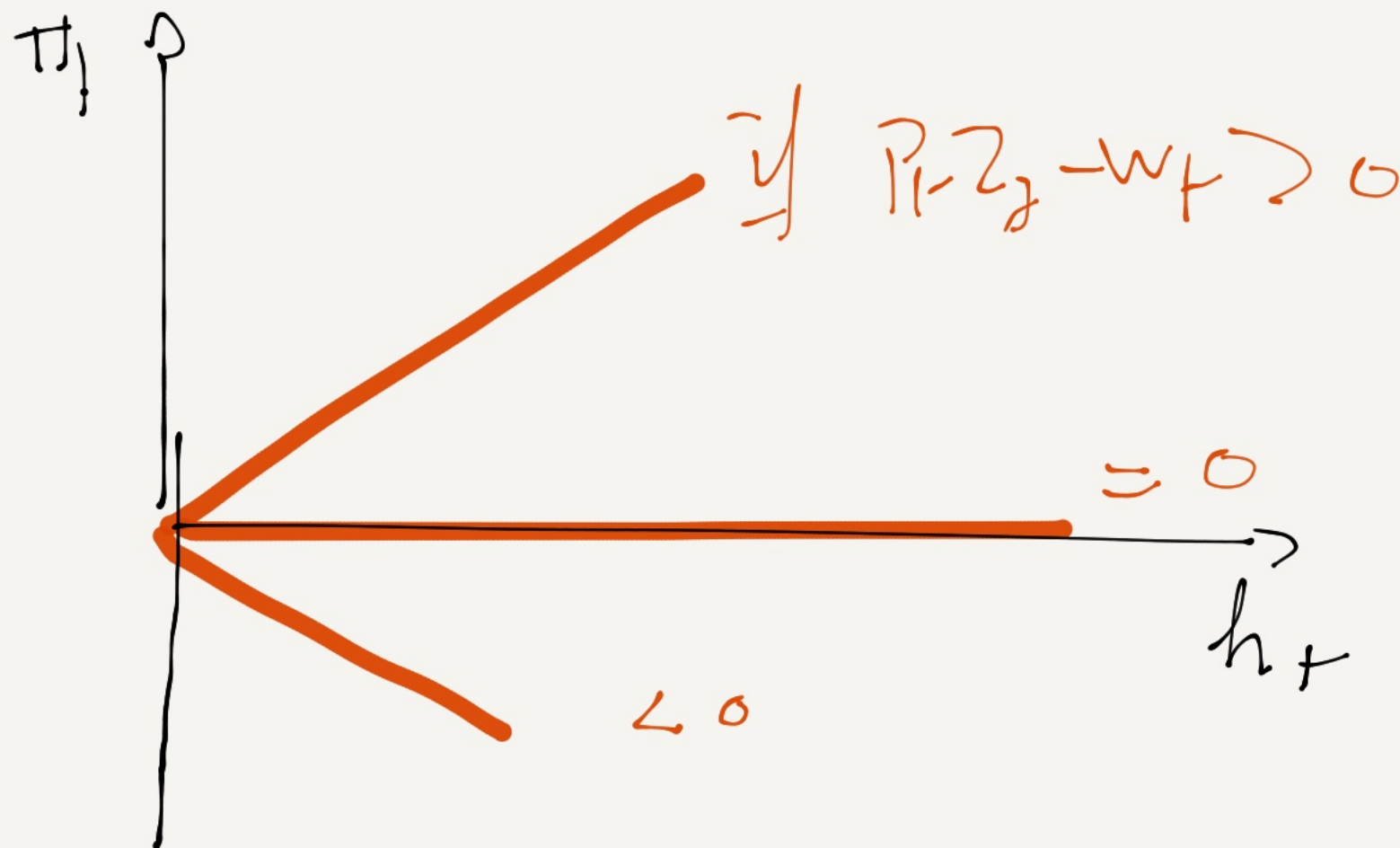
For

$$p_t z_t = w_t \rightarrow \text{ill. written}$$

$\Rightarrow$

$$\pi_t = (p_t z_t - w_t) \times h_t$$

$\hookrightarrow$ Proper FOC



$$\left[\begin{array}{l} \text{if } p_t z_t - w_t > 0, h_t \rightarrow \infty \\ \text{————} = 0 \quad h_t \in \mathbb{R} \\ \text{————} < 0 \quad h_t = 0 \end{array} \right]$$

Household

$$\text{Prof } U = E \sum \beta^t \{ \log (c_t - b c_{t-1}) - h_t \}$$

disutility of
labour

$0 \leq b \leq 1$
(habit persistence)

$$\text{marg. wr. of } c_t = \frac{\partial U}{\partial c_t} = \frac{\beta}{c_t - b c_{t-1}}$$

$$\frac{\partial^2 U}{\partial c_t^2} < 0, \quad \frac{\partial^2 U}{\partial c_{t-1}^2} > 0 \quad | \leftarrow \text{habit persist.}$$

BC

$$P_t C_t = \underbrace{\Pi_t}_{\text{taken as exogenous}} + w_t h_t$$

taken as exogenous

Optimal behavior

$$\max_{\{C_t, h_t\}} E \sum_0^{\infty} \rho^t \left\{ \log(C_t - b C_{t-1}) - h_t \right\} \quad \text{s.t.} \quad R_t C_t = \Pi_t + w_t h_t \quad \forall t$$

↳

$$\max_{\{h_t\}} E \sum_0^{\infty} \rho^t \left\{ \log \left(\frac{\Pi_t + w_t h_t}{P_t} - b \frac{(\Pi_{t-1} + w_{t-1} h_{t-1})}{R_{t-1}} \right) - h_t \right\}$$

FOC in period t

$$\beta^t \left(\frac{w_t / p_t}{c_t - b c_{t+1}} - 1 \right) + \beta^{t+1} E \left(\frac{-w_{t+1} b / p_t}{c_{t+1} - b c_t} \right) = 0$$

$t \rightarrow$

$$\boxed{\frac{1}{c_t - b c_{t+1}} \rightarrow E \left[\frac{\beta b}{c_{t+1} - b c_t} \right] = \left(\frac{w_{t+1}}{p_t} \right)^{-1}}$$

General Equilibrium (Competitive)

$$\{c_t, y_t, h_t^s, h_t^d, w_t, P_t\} \text{ s.t.}$$

① Firms max Π

② Hhs max U

③ Markets clear : $c_t = y_t$
 $h_t^s = h_t^d$

$\Rightarrow$ Walras law : $P_t = 1$

Solving for the G.E.

in eq, $P_t Z_t = W_t \Rightarrow W_t = Z_t$

use the Euler eq,

$$\frac{1}{C_t - b C_{t-1}} - \frac{\beta b}{C_{t+1} - b C_t} = \frac{1}{Z_t} \quad (*)$$

• Initial condition C_0
+ transversality condition

Change of variable: $X_t = \frac{1}{c_t - b c_{t-1}}$

Then (*) $\Rightarrow X_t - \underbrace{\beta b}_t E X_{t+1} = \frac{1}{z_t}$

$$0 < \beta b < 1$$

$$X_t = \frac{1}{z_t} + \beta b E_t (X_{t+1})$$

$$= \frac{1}{z_t} + \beta b E_t \left[\frac{1}{z_{t+1}} + \beta b E_{t+1} X_{t+2} \right]$$

$$= \frac{1}{z_t} + \beta b E_t \left[\frac{1}{z_{t+1}} \right] + (\beta b)^2 E_t X_{t+2}$$

...

$$X_t = \underbrace{\sum_{i=0}^{\infty} (\beta b)^i E_t \left[\frac{1}{Z_{t+i}} \right]}_{\text{soln. in } X_t} + \underbrace{\lim_{i \rightarrow \infty} (\beta b)^i E_t X_{t+i}}_{= 0 \text{ (Transversality)}}$$

[Forward looking]

then

$$X_t = \frac{1}{C_t - b C_{t-1}}$$

$$\Rightarrow \boxed{C_t = \frac{1}{X_t} + b C_{t-1}}$$

C_0 given

LINEAR APPROXIMATION

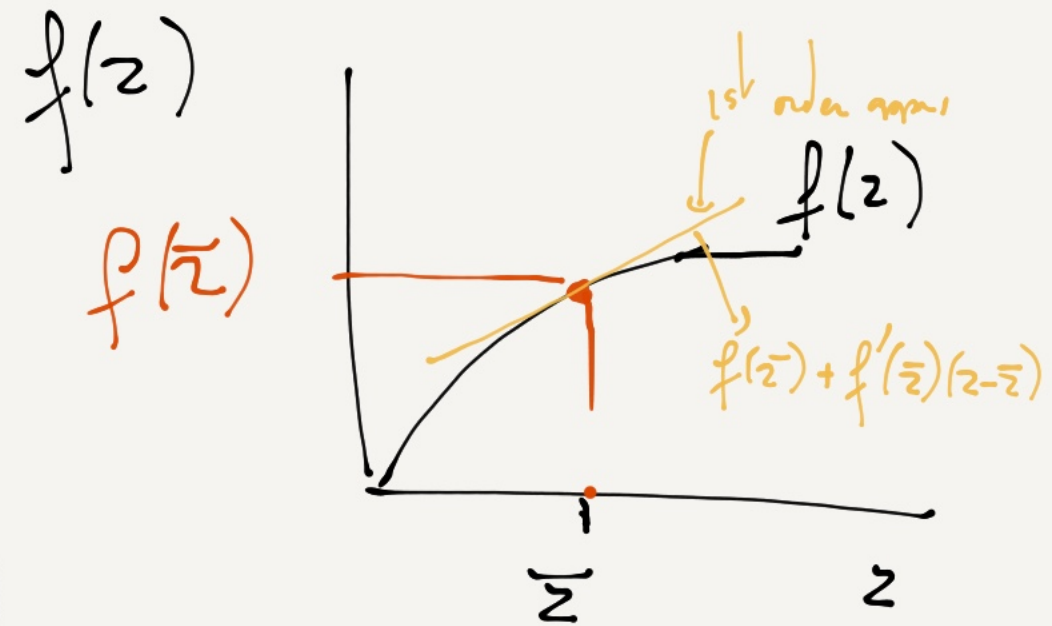
$$y_t = f(x_{1t}, x_{2t})$$

steady state: $\bar{y} = f(\bar{x}_1, \bar{x}_2)$

→ 1st order approx of f around $(\bar{x}_1, \bar{x}_2)$

$$\underbrace{f(x_{1t}, x_{2t})}_{y_t} \approx \underbrace{f(\bar{x}_1, \bar{x}_2)}_{\bar{y}} + f'_1(\bar{x}_1, \bar{x}_2) \times (x_{1t} - \bar{x}_1) + f'_2(\bar{x}_1, \bar{x}_2) \times (x_{2t} - \bar{x}_2)$$

→ $y_t - \bar{y} = f'_1 \times (x_{1t} - \bar{x}_1) + f'_2 \times (x_{2t} - \bar{x}_2)$



$$\underline{y}_t - \underline{y} = \underline{\bar{x}}_1 f'_1 \times \underbrace{(x_{1t} - \bar{x}_1)}_{\bar{x}_1} + \underline{\bar{x}}_2 f'_2 \times \underbrace{(x_{2t} - \bar{x}_2)}_{\bar{x}_2} \quad (\text{A})$$

$$\hat{y}_t = \frac{y_t - \bar{y}}{\bar{y}}, \quad \hat{x}_{1t}, \quad \hat{x}_{2t} \dots$$

$$(A) \quad \underline{y} \hat{y}_t = \bar{x}_1 f'_1 \hat{x}_{1t} + \bar{x}_2 f'_2 \hat{x}_{2t}$$

$$\underline{y} = \bar{f} = f(\bar{x}_1, \bar{x}_2)$$

$$\boxed{\hat{y}_t = \frac{\bar{x}_1 f'_1}{\bar{f}} \hat{x}_{1t} + \frac{\bar{x}_2 f'_2}{\bar{f}} \hat{x}_{2t}}$$

Do the linearization for

$$\left\{ \begin{array}{l} y_t = z_t h_t^2 \\ y_t = c_t + \dot{n}_t \end{array} \right.$$